# Does Compressibility Transfer Across Sensors? A Cross-Dataset Study of CNN Pruning for Optical and SAR Flood Classification

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***Abstract*—Magnitude pruning is the standard route to deploying convolutional networks on the resource-constrained platforms used in disaster response, yet pruning recipes are almost always validated on optical benchmarks. We ask whether the compressibility of a CNN transfers across sensing modalities. Under a single fixed supervised pipeline, we train ResNet18 and VGG-16 flood classifiers on FloodNet (optical UAV imagery) and Sen1Floods11 (Sentinel-1 SAR, converted from per-pixel masks to image-level labels), and prune each model with global unstructured and filter-level structured $L_1$ pruning, fine-tuning every configuration identically; we report final-epoch accuracy, balanced accuracy, macro-F1 and AUC as mean $\pm$ std over three seeds (96 runs). On these two datasets, unstructured compressibility transfers: ResNet18 loses at most $0.03$ macro-F1 at 90% sparsity on both. Structured pruning does not: at 50% filter pruning the FloodNet model loses $0.001$ macro-F1 while the Sen1Floods11 model loses $0.068$, the gap widens monotonically with ratio, and a lower fine-tuning learning rate does not close it. VGG-16 is reported as an instability case: its dense SAR training already fails on one of three seeds (accuracy swings while AUC remains $0.858$, indicating a decision-threshold rather than a representation failure), and its structured variants collapse to constant predictors on both datasets. Our results indicate that sparsity budgets established on one sensor cannot be assumed to hold on another, and that compression choices should be re-validated per dataset and modality.**

***Index Terms*—network pruning, model compression, flood detection, SAR, optical remote sensing, FloodNet, Sen1Floods11**

## I. Introduction

Rapid flood mapping is one of the most time-critical applications of remote sensing: response teams need scene-level situational awareness within hours of a disaster, often from platforms — small UAVs, edge boxes, nanosatellites — that cannot host large networks [1], [4]. Model compression, and in particular magnitude pruning [5], [6], is the standard route to meeting such constraints: weights or whole filters with small magnitude are removed and the network is fine-tuned, frequently recovering accuracy at high sparsity [9].

The evidence base for this practice, however, is narrow in one important way. Pruning methods are overwhelmingly validated on optical (RGB) benchmarks such as ImageNet [12], and when they are applied in remote sensing it is typically to a single dataset at a time. Flood monitoring in practice spans at least two very different sensing modalities: high-resolution optical imagery from UAVs, which is only available in clear conditions, and synthetic aperture radar (SAR), which penetrates cloud cover and is often the only usable source during the flood event itself [2]. A deployment pipeline that compresses a model for the optical case will sooner or later be asked to do the same for the SAR case.

This paper is built around a single question:

> *Does a heavily pruned classic CNN hold its accuracy consistently on both optical and SAR flood imagery, or does one modality break down while the other does not?*

Either answer is informative. If compressibility transfers, then sparsity budgets established on optical benchmarks can be reused for SAR deployments with reasonable confidence. If it does not — if one dataset degrades earlier or collapses outright — then the pruning ratio itself becomes a dataset-dependent hyperparameter, and compression recipes must be re-validated per sensor and dataset.

Answering the question fairly requires removing every confound except the data. We therefore fix a single supervised pipeline — identical architecture definitions, input resolution, augmentation, class balancing, optimizer, schedule, evaluation protocol, pruning operators, sparsity ratios, and fine-tuning budget — and vary *only* the dataset. We instantiate the study with two standard backbones of very different capacity, ResNet18 (11.2M parameters) [10] and VGG-16 (134.3M parameters) [11], on two public flood-classification datasets: FloodNet [1], optical UAV imagery from Hurricane Harvey, and Sen1Floods11 [2], Sentinel-1 SAR chips of global flood events. Sen1Floods11 is labeled per pixel, so we convert its hand-labeled water masks into image-level flooded/non-flooded labels with an explicit, reproducible rule (Section III).

**Contributions.**

- A controlled cross-dataset pruning study (two architectures, two datasets, two pruning families, multiple sparsity ratios, three seeds; 96 trained models) in which the

training pipeline is held fixed and only the dataset varies.
- A reproducible protocol for converting per-pixel SAR flood masks (Sen1Floods11) into image-level classification labels, enabling like-for-like comparison with image-level optical benchmarks (FloodNet).
- Evidence that, on these two datasets, compressibility under *unstructured* pruning transfers (ResNet18: $\leq 0.03$ macro-F1 cost at 90% sparsity on both), whereas *structured* pruning does not (at 50%: $-0.001$ macro-F1 on FloodNet vs. $-0.068$ on Sen1Floods11), and that a lower fine-tuning learning rate does not close the gap — i.e., the SAR-side drop is not an artifact of our fine-tuning recipe.
- An analysis of VGG-16 as an instability case: seed-level failure of dense SAR training that manifests as a decision-threshold (final-epoch) problem while AUC stays stable, and layer-level collapse of its structured variants, which we attribute to the absence of skip connections.

## II. Related Work

### A. Flood Classification Datasets and Baselines

**FloodNet** [1] is a high-resolution ($3000 \times 4000$) UAV imagery dataset collected after Hurricane Harvey, providing a semi-supervised Track-1 challenge with 398 labeled training images (51 flooded, 347 non-flooded), 1,047 unlabeled training images, and validation/test splits whose classification labels were withheld for the challenge evaluation. **Sen1Floods11** [2] provides 4,831 Sentinel-1 SAR chips of $512 \times 512$ pixels covering 11 global flood events, including 446 hand-labeled chips whose per-pixel masks distinguish water, non-water, and invalid pixels; it is designed for per-pixel flood segmentation rather than image-level classification, which motivates the label conversion of Section III.

On the modeling side, Khose *et al.* [3] established a strong FloodNet Track-1 classification baseline: a ResNet18 trained with pseudo-label semi-supervision [13] reaching 96.70% test accuracy with 11.6M parameters — outperforming substantially larger InceptionV3, ResNet50 and Xception baselines. Their result motivates our choice of ResNet18 as the small reference backbone; in this work, however, we deliberately use *purely supervised* training for all runs, since pseudo-labeling would entangle compression effects with semi-supervision effects. Closest to our deployment motivation, Khan *et al.* [4] propose Flood-ResNet50, a modified ResNet50 for flooded/non-flooded UAV image classification evaluated on-device (Jetson Nano) with 96.43% accuracy and 86.36% F1 on FloodNet. Their study targets efficiency via architecture design on a single optical dataset; ours is complementary, asking whether a *given* architecture compressed by pruning behaves consistently across an optical and a SAR dataset.

### B. Network Pruning

Magnitude pruning removes parameters deemed unimportant by weight magnitude. *Unstructured* pruning [5] zeroes individual weights globally across layers and achieves high sparsity with little accuracy loss, though realized speedups require sparse kernels. *Structured* pruning [6] removes entire filters or neurons, producing dense sub-networks that accelerate on commodity hardware but constraining what can be removed. Subsequent work has questioned what pruning actually preserves: the lottery-ticket hypothesis [7] locates sparse trainable sub-networks, Liu *et al.* [8] argue that the value of pruning lies largely in the architecture it induces, and Blalock *et al.* [9] document that pruning papers overwhelmingly evaluate on a small set of optical benchmarks under inconsistent protocols. To our knowledge, no prior work has held a flood-classification pipeline fixed and measured how the *same* pruning protocol behaves across an optical and a SAR dataset; this is the gap we address.

## III. Experimental Setup

### A. Datasets and Label Conversion

**FloodNet (optical).** We use the 398 labeled Track-1 training images (51 flooded / 347 non-flooded). Because classification labels for the official validation and test splits were withheld for the challenge server, all labeled evaluation must come from these 398 images: we use a stratified 50/50 split (199 train / 199 validation, fixed seed), mirroring the split strategy of the public baseline implementation [3]. Images are resized from $3000 \times 4000$ to $400 \times 300$.

**Sen1Floods11 (SAR).** We use the hand-labeled subset: Sentinel-1 chips with two polarization bands (VV, VH) and per-pixel masks with values $\{1$=water, $0$=non-water, $-1$=invalid$\}$. Since the dataset is natively per-pixel, we convert each chip to an image-level label with the rule

$$y = \mathbf{1}\left[\frac{|\{p : m_p = 1\}|}{|\{p : m_p \neq -1\}|} \;\geq\; \tau\right], \qquad \tau = 0.01, \tag{1}$$

i.e., a chip is *flooded* if at least 1% of its valid pixels are water; chips with no valid pixels are discarded. This yields 441 usable chips (293 flooded / 148 non-flooded), split 50/50 stratified into 220 train / 221 validation chips with the same fixed seed. For input compatibility with ImageNet-pretrained backbones, SAR backscatter is clipped to $[-50, 1]$ dB, min–max scaled to $[0, 1]$, and stacked as three channels $(\mathrm{VV}, \mathrm{VH}, (\mathrm{VV}{+}\mathrm{VH})/2)$, then bilinearly resized to $400 \times 300$.

The two datasets are imbalanced in *opposite* directions: flooded images are the minority class in FloodNet (12.8%) but the majority class in Sen1Floods11 (66.4%). We therefore report balanced accuracy and macro-F1 alongside accuracy and AUC, and we compare each pruned model only to its own dense baseline, never across datasets in absolute terms.

### B. Unified Training Pipeline

All dense baselines and all fine-tuning runs share one pipeline, identical across datasets and architectures: ImageNet-pretrained backbone [12] with the final layer replaced by a 2-logit head; cross-entropy loss; random horizontal and vertical flips as the only augmentation; a class-balanced weighted random sampler; SGD with learning rate $0.01$ and no momentum; and 50 epochs. We deliberately evaluate the *final-epoch* checkpoint — no early stopping and no best-validation

TABLE I
RESNET18, FINAL-EPOCH MACRO-F1 AND AUC (MEAN ± STD, 3 SEEDS). PARAMS = NONZERO PARAMETERS (M). FT LR = 0.01.

| | | | FloodNet | | Sen1Floods11 | |
|---|---|---|---|---|---|---|
| **Method** | **Ratio** | **Par.** | **mF1** | **AUC** | **mF1** | **AUC** |
| dense | 0% | 11.18 | .929±.011 | .991±.003 | .827±.009 | .908±.013 |
| unstruct. | 50% | 5.59 | .917±.010 | .995±.002 | .833±.006 | .912±.011 |
| unstruct. | 90% | 1.13 | .911±.009 | .989±.002 | .799±.013 | .888±.017 |
| structured | 30% | 7.82 | .925±.012 | .986±.012 | .768±.030 | .858±.015 |
| structured | 50% | 5.59 | .928±.016 | .989±.004 | .759±.006 | .841±.010 |
| structured | 70% | 3.37 | .857±.063 | .930±.052 | .734±.028 | .834±.013 |
| structured | 90% | 1.13 | .798±.064 | .930±.018 | .577±.038 | .763±.026 |

selection — so that reported numbers are not inflated by checkpoint cherry-picking. Each configuration is run with three seeds (42, 124, 2024) that vary initialization, sampling and augmentation randomness while the data split stays fixed; we report mean ± std. The only architecture-forced difference is the batch size (64 for ResNet18, 16 for VGG-16, due to memory at $400 \times 300$); it is held constant across datasets so that every FloodNet-vs-Sen1Floods11 comparison is batch-size matched.

### C. Pruning Protocol

Starting from the dense checkpoint of the *same seed* we apply, separately:

- **Global unstructured** $L_1$ [5]: individual weights of all convolutional and linear layers are ranked globally by magnitude and the smallest fraction $r$ is zeroed, with $r \in \{0.5, 0.9\}$.
- **Structured** $L_1$ [6]: for every convolutional layer with more than 8 output channels, the fraction $r$ of output filters with smallest $L_1$ norm is masked; for every linear layer with more than 8 output units, the same rule is applied to rows; $r \in \{0.3, 0.5, 0.7, 0.9\}$. The final 2-unit classifier is never pruned.

Each pruned model is fine-tuned for 50 epochs with the dense recipe (SGD, lr $0.01$), and the final-epoch model is evaluated. As a control for the fine-tuning recipe, we additionally repeat a subset of structured configurations (ResNet18 at 50% and 90%, VGG-16 at 70%, both datasets, two seeds) with fine-tuning learning rate $0.001$. Sparsity masks are kept fixed during fine-tuning, and we report nonzero parameter counts. In total: 12 dense runs, 48 structured runs, 24 unstructured runs, and 12 learning-rate-control runs — 96 trained models.

## IV. RESULTS AND DISCUSSION

### A. ResNet18: Unstructured Compressibility Transfers; Structured Does Not

Table I and Fig. 1 contain the main result. Under **unstructured** pruning, ResNet18 is compressible to 90% sparsity (1.13M nonzero parameters) at a macro-F1 cost of at most $0.03$ on *both* datasets: $0.929 \rightarrow 0.911$ on FloodNet and $0.827 \rightarrow 0.799$ on Sen1Floods11, with AUC essentially unchanged. On these two datasets, unstructured compressibility transfers.

TABLE II
FINE-TUNING LEARNING-RATE CONTROL: RESNET18 STRUCTURED PRUNING, MACRO-F1 (MEAN ± STD; LR 0.01: 3 SEEDS, LR 0.001: 2 SEEDS). THE LOWER RATE DOES NOT RECOVER THE SEN1FLOODS11 DROP.

| **Dataset** | **Ratio** | **mF1 (lr .01)** | **mF1 (lr .001)** |
|---|---|---|---|
| FloodNet | 50% | .928±.016 | .914±.009 |
| FloodNet | 90% | .798±.064 | .699±.093 |
| Sen1Floods11 | 50% | .759±.006 | .718±.010 |
| Sen1Floods11 | 90% | .577±.038 | .609±.013 |

TABLE III
VGG-16, FINAL-EPOCH MACRO-F1 AND AUC (MEAN ± STD, 3 SEEDS; LR-CONTROL ROW: 2 SEEDS). †: AT LEAST ONE SEED COLLAPSED TO A CONSTANT PREDICTOR (BALANCED ACCURACY 0.5).

| | | FloodNet | | Sen1Floods11 | |
|---|---|---|---|---|---|
| **Method** | **Ratio** | **mF1** | **AUC** | **mF1** | **AUC** |
| dense | 0% | .924±.006 | .990±.000 | .621±.262† | .858±.003 |
| unstruct. | 50% | .924±.006 | .990±.001 | .601±.248† | .849±.005 |
| unstruct. | 90% | .943±.015 | .980±.020 | .597±.245† | .846±.014 |
| struct. | 30% | .386±.199† | .737±.035 | .586±.237† | .834±.005 |
| struct. | 50% | .803±.076 | .940±.027 | .519±.190† | .788±.007 |
| struct. | 70% | .621±.064 | .675±.187 | .300±.070† | .502±.225 |
| struct. | 90% | .230±.167† | .500±.000 | .300±.070† | .500±.000 |
| str. lr .001 | 70% | .289±.177† | .922±.003 | .325±.074† | .700±.000 |

Under **structured** pruning the picture changes. On FloodNet, filter pruning is free up to 50% ($0.929 \rightarrow 0.928$ macro-F1, $-0.001$) and only degrades at 70–90%. On Sen1Floods11 the same operator costs macro-F1 from the very first ratio: $-0.058$ at 30%, $-0.068$ at 50%, $-0.092$ at 70%, and $-0.249$ at 90%, with AUC falling in step ($0.908 \rightarrow 0.763$). The degradation is monotonic, exceeds seed noise by an order of magnitude at 50% (std $0.006$), and has no counterpart on the optical dataset at the same ratios. We state this as a finding *on these two datasets*: the same structured pruning protocol that is harmless at 50% on FloodNet removes capacity that the Sen1Floods11 task evidently needs.

### B. The Learning-Rate Control

One mundane explanation would be that our fine-tuning recipe (SGD, lr $0.01$, 50 epochs) is simply mis-tuned for pruned SAR models. Table II tests this with fine-tuning at lr $0.001$. The lower rate does not recover Sen1Floods11: $0.718$ vs. $0.759$ at 50% (worse), $0.609$ vs. $0.577$ at 90% (marginally better, still $0.218$ below dense). On FloodNet it is neutral at 50% and harmful at 90%. The structured-pruning gap between the two datasets is therefore not an artifact of the fine-tuning learning rate.

### C. VGG-16: An Instability Case

We report VGG-16 not as a second clean data point but as an instability case (Table III). Three observations.

First, dense VGG-16 training on Sen1Floods11 is itself unreliable: accuracy is $0.661 \pm 0.230$ over three seeds. Table IV breaks this down per seed: seeds 124 and 2024 train normally

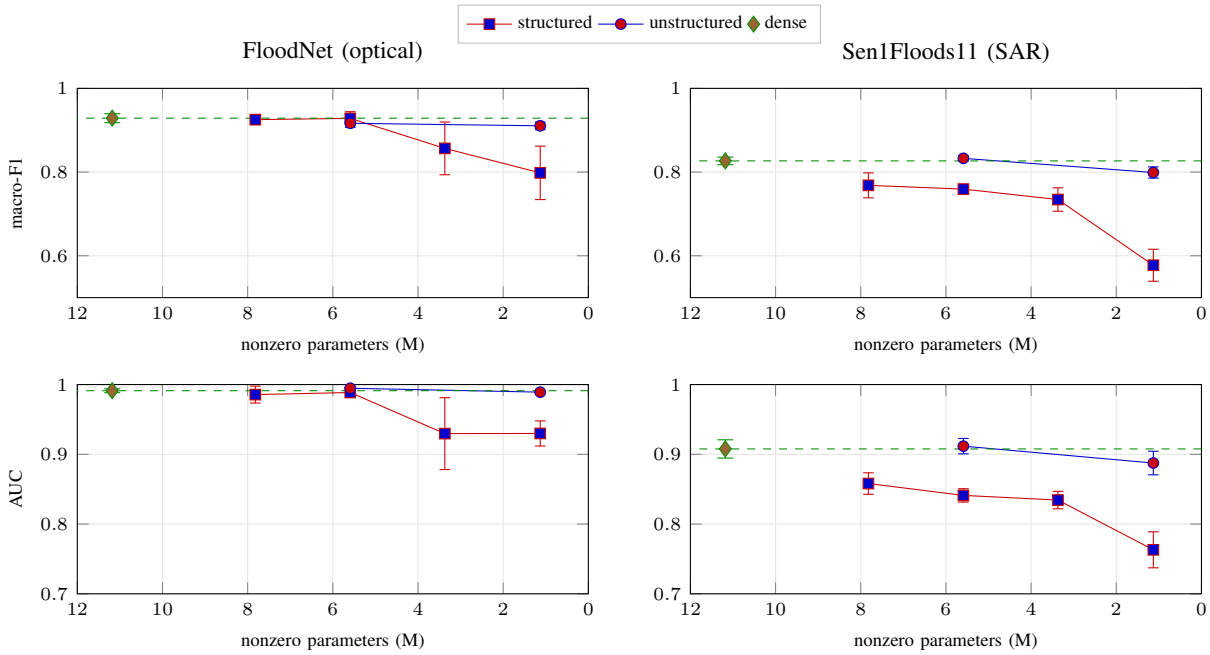



Fig. 1. ResNet18 under pruning: macro-F1 (top) and AUC (bottom) vs. nonzero parameters, on FloodNet (left) and Sen1Floods11 (right). Error bars: std over 3 seeds; dashed line: dense baseline. Unstructured pruning tracks the dense baseline on both datasets down to 1.13M parameters; structured pruning tracks it on FloodNet up to 50% but degrades monotonically on Sen1Floods11 from the first ratio.

TABLE IV
PER-SEED VGG-16 *dense* RESULTS ON SEN1FLOODS11 (FINAL EPOCH). SEED 42 COLLAPSES TO THE CONSTANT NON-FLOODED PREDICTOR, YET ITS AUC REMAINS 0.858: THE LEARNED RANKING IS INTACT AND THE FAILURE IS THE FINAL-EPOCH DECISION THRESHOLD.

| Seed | Acc | Bal. Acc | mF1 | AUC |
|---|---|---|---|---|
| 42 (failed) | 0.335 | 0.500 | 0.251 | 0.858 |
| 124 | 0.819 | 0.800 | 0.798 | 0.861 |
| 2024 | 0.828 | 0.827 | 0.814 | 0.855 |

(macro-F1 $\approx$ 0.80), while seed 42 ends at the constant non-flooded predictor (balanced accuracy 0.500). Crucially, the failed seed's AUC is 0.858 — indistinguishable from the healthy seeds. The representation ranks flooded above non-flooded chips correctly; what fails is the position of the decision boundary at the final epoch. This is a final-epoch thresholding pathology of a high-capacity model on 220 training chips, not evidence that VGG-16 cannot encode the SAR task.

Second, structured VGG-16 collapses on *both* datasets (dagger rows: 90% on FloodNet, 70–90% on Sen1Floods11, and sporadically at 30% on FloodNet), always to a constant predictor with AUC at or near 0.5. A layer-level count explains why: our per-layer operator removes the $\mathrm{round}(r\,n)$ smallest-norm filters from a layer of width $n$, so at $r = 0.9$ the VGG-16 stem retains only 6 of 64 filters (13/128, 26/256, 51/512 deeper on, 410/4096 in the fully connected layers). Because VGG-16 is a single serial path with no skip connections, every prediction must pass through these near-empty layers, whereas ResNet18's parameter-free identity shortcuts — which are never pruned — preserve an information path around damaged blocks, which is why ResNet18 degrades gracefully where VGG-16 collapses.

Third, VGG-16 at 90% *unstructured* on FloodNet reaches macro-F1 $0.943 \pm 0.015$ vs. dense $0.924 \pm 0.006$; we note explicitly that this difference is within seed noise and we do *not* claim pruning improves VGG-16 — the supportable statement is that 90% unstructured sparsity is cost-free for VGG-16 on FloodNet.

### D. Answer to the Title Question

On these two datasets: compressibility transfers under unstructured magnitude pruning — a 90%-sparse ResNet18 is within 0.03 macro-F1 of dense on both the optical and the SAR task — but it does not transfer under structured filter pruning, where the optical task tolerates 50% filter removal at no cost while the SAR task pays 0.068 macro-F1 at the same ratio and degrades monotonically thereafter, robustly to the fine-tuning learning rate. Practically: a sparsity budget validated on one sensor's data cannot be assumed to hold for another's, and structured compression in particular should be re-validated, with balanced metrics and AUC, on every target dataset.

## V. LIMITATIONS

**Two datasets, many differences.** FloodNet and Sen1Floods11 differ not only in sensor physics but also in size (398 vs. 441 labeled images), spatial resolution ($3000 \times 4000$ UAV frames vs. $512 \times 512$ 10 m satellite chips), class imbalance (12.8% vs. 66.4% flooded), scene content

and geography, and label provenance (direct image-level annotation vs. labels derived from segmentation masks via Eq. (1)). Our design holds the pipeline fixed and can therefore attribute the gap to *the dataset*, but not to sensor modality specifically. All claims in this paper are scoped to these two datasets; generalizing to "SAR vs. optical" requires paired acquisitions and more datasets per modality.

**Three seeds, small evaluation sets.** Mean $\pm$ std over three seeds distinguishes the large structured-pruning effects (e.g., the $0.068$ drop at 50% against std $0.006$) but not fine-grained differences; evaluation uses 199 images (only $\sim$25 flooded) on FloodNet and 221 chips on Sen1Floods11, so single-image changes are visible in the metrics. The withheld FloodNet val/test labels force evaluation on a split of the 398 labeled images.

**Final-epoch protocol.** Reporting the final epoch avoids checkpoint cherry-picking but exposes threshold instabilities (the VGG-16 seed-42 case); a calibration step or threshold selection on a held-out set would likely remove that failure mode and should accompany deployments.

**Label conversion threshold.** Sen1Floods11 image labels depend on the $\tau = 1\%$ rule of Eq. (1); a sensitivity analysis over $\tau$ is left to future work.

**Masked rather than physically removed structure.** Pruning is applied as weight masks; structured channels are zeroed, not excised, so FLOP reductions are theoretical, and unstructured sparsity yields no wall-clock speedup without sparse kernels. Neither affects the accuracy comparison, which is our focus.

**ImageNet pretraining.** Both backbones start from RGB-pretrained weights, which is standard but favorable to the optical dataset; SAR-specific pretraining might shift where structured pruning begins to hurt.

## VI. Conclusion

We asked whether CNN compressibility transfers across sensing modalities and answered it with a controlled 96-run study on FloodNet (optical) and Sen1Floods11 (SAR) under a single fixed pipeline. On these two datasets, unstructured magnitude pruning transfers — ResNet18 keeps macro-F1 within $0.03$ of dense at 90% sparsity on both — while structured filter pruning does not: harmless at 50% on the optical dataset, it costs $0.068$ macro-F1 on the SAR dataset at the same ratio, degrades monotonically with ratio, and is not repaired by a lower fine-tuning learning rate. VGG-16 serves as a cautionary instability case: seed-level dense failures on SAR that are threshold rather than representation pathologies (AUC stable at $0.858$), and structured collapse on both datasets traceable to near-empty layers in a skip-connection-free architecture. The practical recommendation is direct: compression choices — especially structured ones — should be re-validated per dataset with balanced accuracy, macro-F1 and AUC reported together, and sparsity budgets should not be ported across sensors untested. Future work includes paired optical/SAR acquisitions to isolate modality, more seeds and datasets, sensitivity analysis of the label-conversion threshold, and channel-importance analysis of which filters structured pruning removes on SAR.

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